Writing Software User Guides

HABILIDADES PROFESIONALES PARA INGENIEROS (HPI)

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Introduction to User Guides

- Definition: Documents that help users understand and operate software
- Types: User Interface (UI) guides and Integration guides
- Importance: Clear communication ensures better user experience

Purpose of a UI Guide

- Helps users navigate and operate the software interface
- Explains features, functions, and settings
- Often used for non-technical end users

Structure of a UI Guide

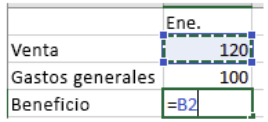
- Introduction to the software
- Detailed feature explanations
- Step-by-step instructions

Crear una fórmula que haga referencia a valores de otras celdas

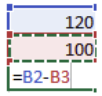
1. Seleccione una celda.
2. Escriba el signo igual =.

   Nota: Las fórmulas de Excel siempre comienzan con el signo igual.

3. Seleccione una celda o escriba su dirección en la celda seleccionada.



4. Escriba un operador. Por ejemplo, "-" para restar.
5. Seleccione la celda siguiente o escriba su dirección en la celda seleccionada.



6. Presione Entrar. El resultado del cálculo se mostrará en la celda que contenga la fórmula.

Example: Elements of a UI Guide

- Screenshots and diagrams to explain features
- Instructions for each feature or button
- Troubleshooting common issues

Purpose of an Integration Guide

- Helps developers integrate software libraries or APIs
- Target audience: Technical users (developers, engineers)
- Focus on code examples, setup, and configuration

Structure of an Integration Guide

- Overview of the library or API
- Installation and setup instructions
- Code examples for integration

Example: https://github.com/google/gson

Example: Elements of an Integration Guide

- Prerequisites and environment setup
- Code snippets with explanations
- Error handling and troubleshooting

Maven:

```
<dependency>
  <groupId>com.google.code.gson</groupId>
  <artifactId>gson</artifactId>
  <version>2.11.0</version>
</dependency>
```

Gson ja

Build

Requi

Minimu

- Gs
- Gs
- Gs

Documentation

- API Javadoc: Documentation for the current release
- User guide: This guide contains examples on how to use Gson in your code
- Troubleshooting guide: Describes how to solve common issues when using Gson
- Releases and change log: Latest releases and changes in these versions; for older relea `CHANGELOG.md`
- Design document: This document discusses issues we faced while designing Gson. It a comparison of Gson with other Java libraries that can be used for Json conversion

Common Technologies for Writing User Guides

- Markdown: Lightweight markup language
- Wikis: Collaborative documentation (e.g., Confluence, GitHub Wiki)
- Code documentation (e.g., javadoc)

UI Guides vs. Integration Guides

- UI Guides:
  - For end-users
  - Focus on interface and functionalities
- Integration Guides:
  - For developers
  - Focus on code, setup, and APIs

Best Practices for Writing User Guides

USE CLEAR, CONCISE LANGUAGE

INCLUDE VISUALS (SCREENSHOTS, DIAGRAMS)

ORGANIZE INFORMATION LOGICALLY

Visual Elements in User Guides

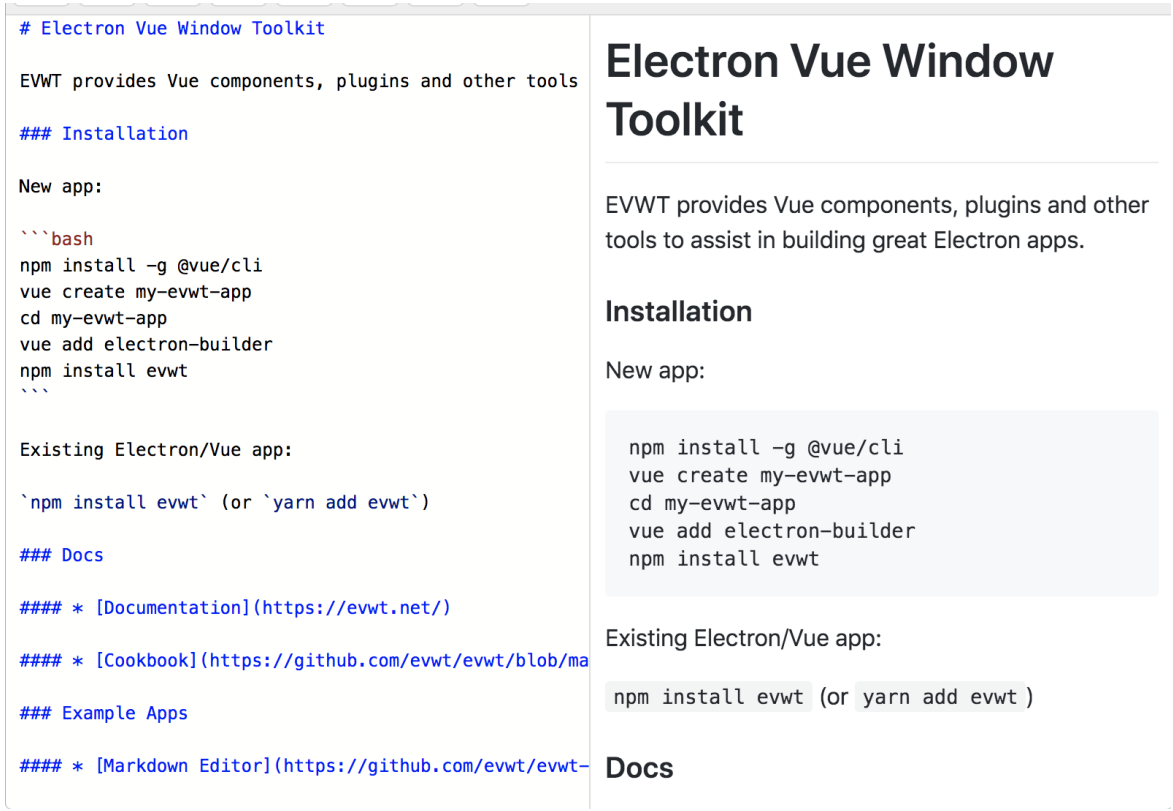
- Importance of screenshots and annotations
- Diagrams to explain processes or flows
- Use of videos or GIFs to enhance learning

Choosing the Right Format for Your Guide

- Markdown: Simple, easy to format, used in GitHub/Wikis
- Javadoc: For Java libraries and APIs
- Wikis: Collaborative documentation for teams

When to Use Markdown

- Ideal for lightweight documentation
- Commonly used in GitHub repositories
- Easy to learn and maintain



```
# Electron Vue Window Toolkit

EVWT provides Vue components, plugins and other tools

### Installation

New app:

```bash
npm install -g @vue/cli
vue create my-evwt-app
cd my-evwt-app
vue add electron-builder
npm install evwt
```

Existing Electron/Vue app:

`npm install evwt` (or `yarn add evwt`)

Docs

* [Documentation](https://evwt.net/)

* [Cookbook](https://github.com/evwt/evwt/blob/ma

Example Apps

* [Markdown Editor](https://github.com/evwt/evwt-
```

Electron Vue Window Toolkit

EVWT provides Vue components, plugins and other tools to assist in building great Electron apps.

Installation

New app:

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vue create my-evwt-app
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Docs

When to Use Code Documentation

- Java doc is great for documenting Java code
- Generates API documentation automatically
- Integrates well with Java IDEs

Overview Package Class Use Tree Deprecated Index Help

Java™ Platform Standard Ed. 7

Prev Class Next Class Frames No Frames All Classes

Summary: Nested | Field | Constr | Method Detail: Field | Constr | Method

java.util

Class Scanner

java.lang.Object
  java.util.Scanner

All Implemented Interfaces:

Closeable, AutoCloseable, Iterator<String>

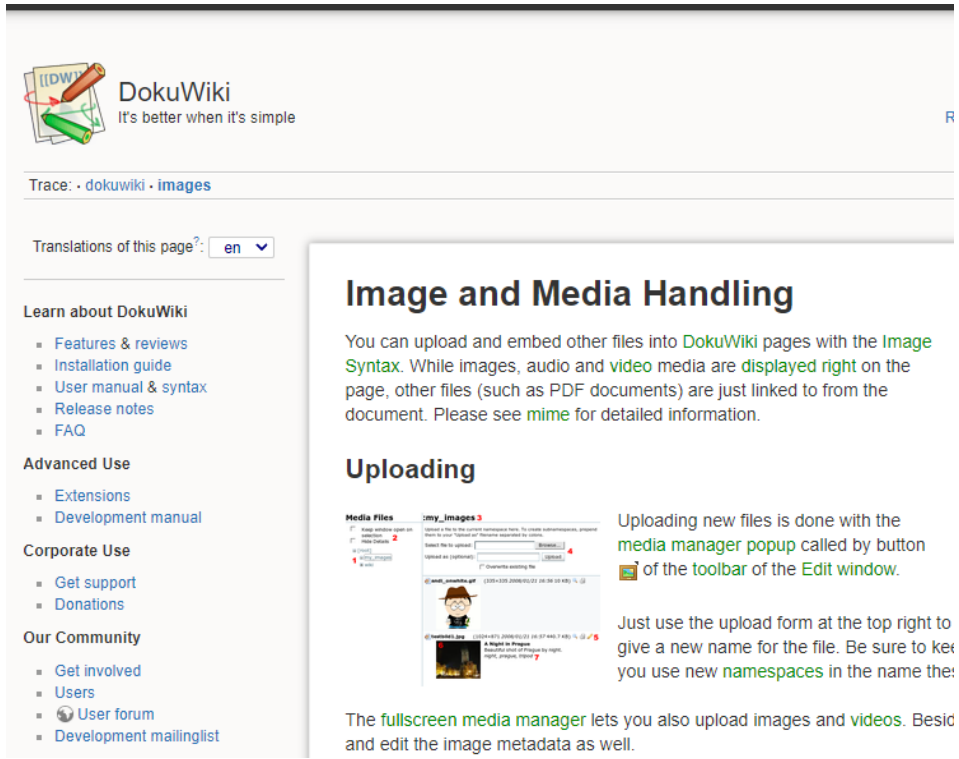
```
public final class Scanner
extends Object
implements Iterator<String>, Closeable
```

A simple text scanner which can parse primitive types and strings using regular expressions.

A `Scanner` breaks its input into tokens using a delimiter pattern, which by default matches whitespace. The resulting tokens may then be converted into values of different types using the various `next` methods.

When to Use Wikis

- Great for collaborative environments
- Allows multiple users to contribute and edit
- Popular tools: Confluence, GitHub Wiki, Notion



Organizing a User Guide

- Table of Contents for easy navigation
- Separate sections for different topics or features
- Use of headers and sub-headers to structure content

Testing and Updating User Guides

- Test the guide with actual users
- Regularly update as the software evolves
- Get feedback from both technical and non-technical users

Javadoc
Python documentation
Wiki
Markdown

What is Javadoc?

- A tool for generating API documentation from Java code
- Extracts documentation from comments in the source code
- Standard for documenting Java projects

Key Features of Javadoc

- Automatically generates HTML documentation
- Supports annotations like `@param`, `@return`, and `@see`
- Integration with Java IDEs for real-time documentation

Javadoc Example

```java
/**
 * Calculates the sum of two numbers.
 * @param a First number
 * @param b Second number
 * @return Sum of a and b
 */
public int sum(int a, int b) {
    return a + b;
}
```

```
© Main

public int sum(
    int a,
    int b
)

Calculates the sum of two numbers.

Params:  a – First number
         b – Second number

Returns: Sum of a and b
```

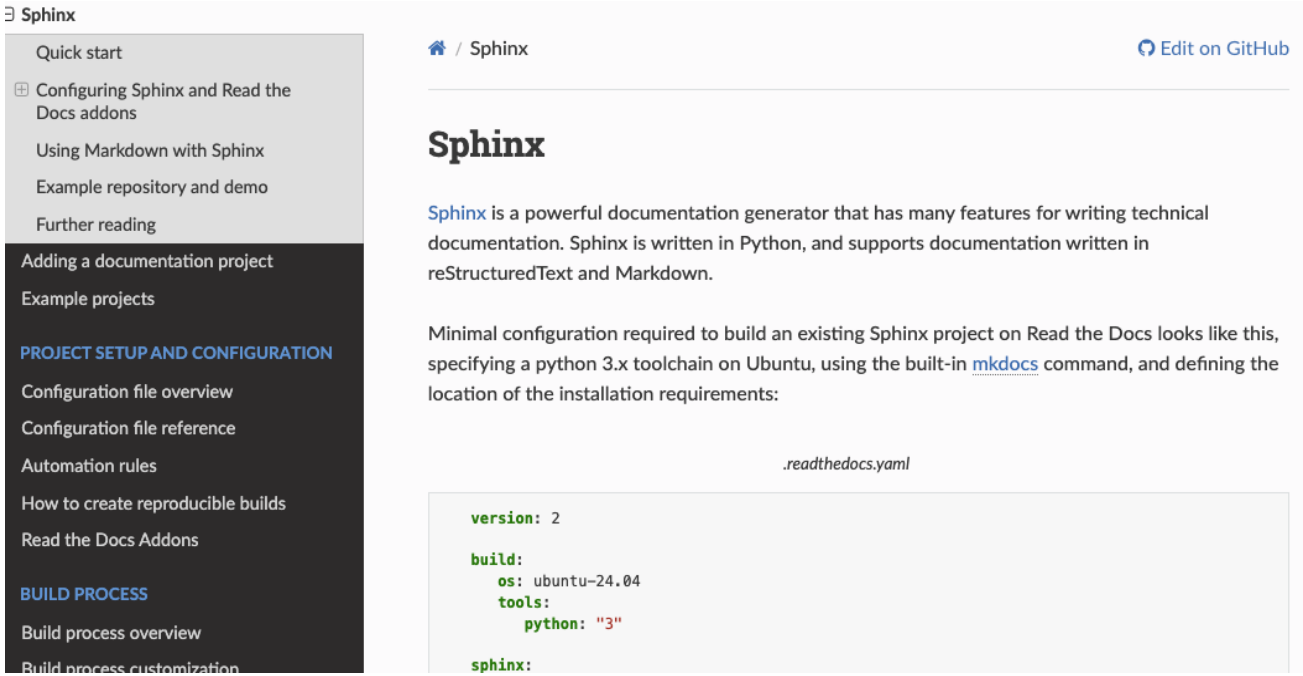
Python Docstring Syntax

- Docstrings: Special comments used for documentation
- Placed at the beginning of functions, classes, and modules

```
def function_name(param1, param2):
    """
    Brief description of the function.
    def function_name(parami: Any,
                      param2: Any) -> Any
    Brief description of the function.
    Args: parami (type): Description of parami.
    param2 (type): Description of param2.
    Returns: type: Description of the return value.
        type: Description of the return value.
    """
    return result
```

Tools for Python Documentation

- **Sphinx**: Generates HTML documentation from docstrings
- **pdoc3**: Lightweight tool for documenting Python libraries
- **reStructuredText (RST)**: A format often used with Sphinx



What is Markdown?

- A lightweight markup language for formatting text
- Used in GitHub repositories, wikis, and documentation
- Easy to write and read in plain text

Basic Markdown Syntax

- Headers: `# Header 1`, `## Header 2`, `### Header 3`
- Bold: `**Bold text**`
- Italics: `*Italic text*`
- Lists:
- Unordered: `* Item`
- Ordered: `1. Item`

Markdown for Code Blocks

- Inline code: `Here is some inline code`
- **Code blocks:** ```` def hello_world(): print("Hello, World!") ```
- Syntax highlighting for multiple languages

Markdown for Links and Images

- Links: `[Text](URL)`
- Images: `![Alt Text](Image URL)`
- Supports embedding of images directly in documentation

Tools and Platforms Using Markdown

- **GitHub:** For README files and wikis
- **Jupyter Notebooks:** Markdown for text cells
- **Static Site Generators:** Like Jekyll or Hugo

Activities

- Introduction to markdown